

Port Security: -

Port Security is a feature available on Cisco switches that restricts input to an interface by limiting and identifying MAC addresses of the devices that are allowed to access the port. It helps prevent unauthorized access and mitigates security threats like MAC address spoofing and CAM table overflow attacks.

Key Features of Port Security

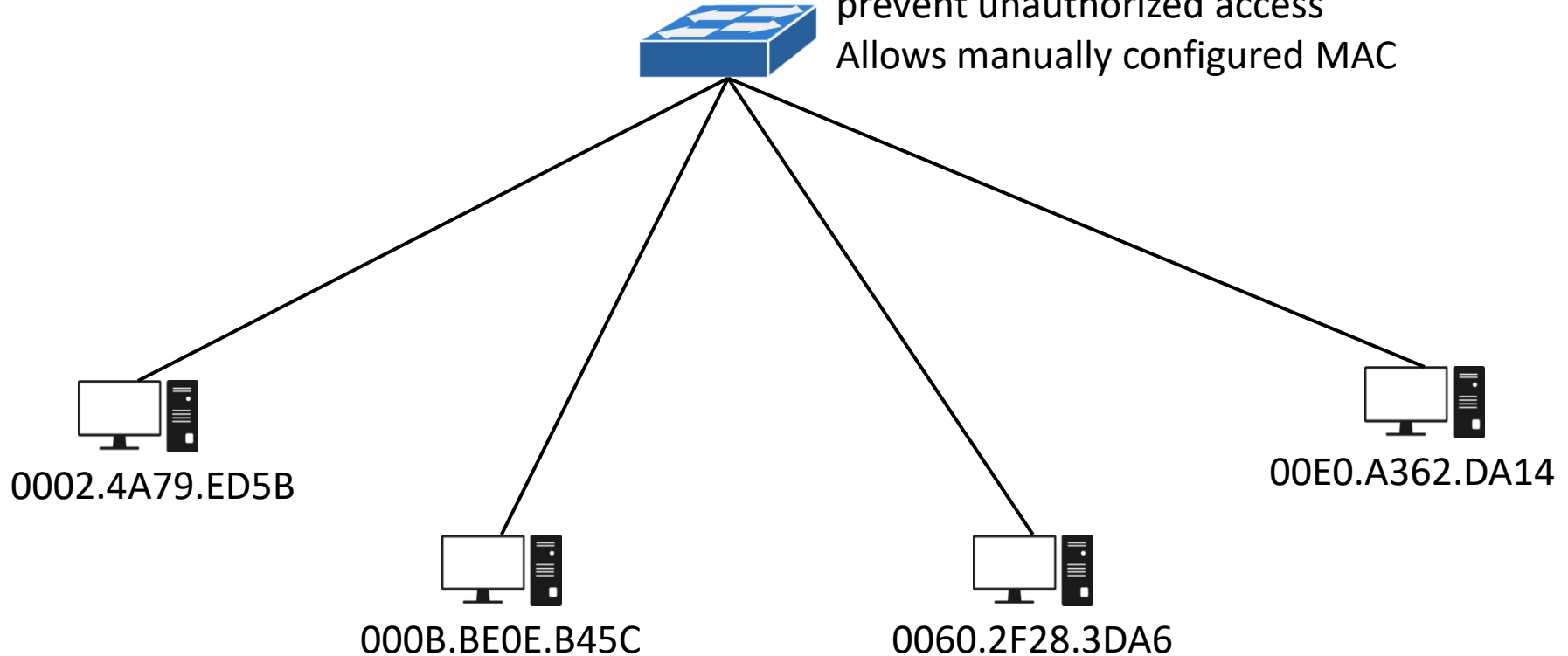
Limits MAC Addresses: Restricts the number of MAC addresses that can connect to a port.

Static or Dynamic Learning: Allows manually configured MAC addresses or dynamically learned ones.

Violation Modes: Determines the action taken when an unauthorized MAC address is detected.

Port Security: -

Identifying MAC addresses of the devices
prevent unauthorized access
Allows manually configured MAC



Port Security Configuration Steps: -

1. Enable Port Security

Port security is applied on access ports. First, set the interface as an access port:

Switch(config)# interface FastEthernet 0/1

Switch(config-if)# switchport mode access

Switch(config-if)# switchport port-security

2. Set Maximum MAC Addresses

Define how many MAC addresses can be learned dynamically on the port:

Switch(config-if)# switchport port-security maximum 2

Port Security Configuration Steps: -

3. Configure Allowed MAC Addresses (Optional)

Manually assign MAC addresses that are allowed:

```
Switch(config-if)# switchport port-security mac-address  
0001.2233.4455
```

```
Switch(config-if)# switchport port-security mac-address  
0066.7788.9900
```

Alternatively, dynamically learned addresses can be stored in the running configuration using **sticky learning**:

```
Switch(config-if)# switchport port-security mac-address  
sticky
```

Port Security Configuration Steps: -

4. Set Violation Mode

When an unauthorized device tries to access the port, the switch can take different actions:

Protect: Drops unauthorized traffic without logging.

Restrict: Drops unauthorized traffic and logs violations.

Shutdown (default): Disables the port (Err-disabled mode).

Switch(config-if)# switchport port-security violation shutdown

Other options:

Switch(config-if)# switchport port-security violation restrict

Switch(config-if)# switchport port-security violation protect

Port Security Configuration Steps: -

5. Verify Port Security Configuration

To check the status and violations:

Switch# show port-security interface FastEthernet 0/1

Switch# show port-security address

6. Recover from Port Shutdown

If a port is shut down due to a violation, manually bring it back:

Switch(config)# interface FastEthernet 0/1

Switch(config-if)# shutdown

Switch(config-if)# no shutdown

Where to use port Security: -

Use Cases of Port Security

- Prevents unauthorized devices from accessing the network.
- Mitigates CAM table overflow attacks.
- Ensures only known devices connect to specific switch ports.